

ANTANANARIVO IVATO, Madagascar

WMO: 670830

Lat: 18.797S

Lon: 47.479E

Elev: 1280

StdP: 86.87

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.0	9.0	4.9	6.3	19.6	6.1	6.8	17.1	8.2	17.3	7.5	17.3	0.8	160	0.345

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	10.9	30.0	19.5	29.1	19.7	28.2	19.7	23.2	26.6	22.5	26.1	21.7	25.6	2.9	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.1	19.7	25.0	21.2	18.6	24.1	20.8	18.2	23.7	76.6	26.5	73.4	26.1	70.5	25.6	26.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.4	6.7	6.1	DB	5.2	32.2	1.1	0.8	4.4	32.8	3.8	33.2	3.2	33.7	2.3	34.2	
			WB	5.0	25.0	1.2	1.2	4.2	25.9	3.5	26.6	2.8	27.2	2.0	28.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	19.6	22.2	22.1	21.9	20.7	18.6	16.2	15.4	16.2	18.0	20.3	21.6	22.4
	DBStd	2.95	1.18	1.21	1.25	1.38	1.66	1.65	1.53	1.59	1.89	1.94	1.62	1.21
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	279	0	0	0	1	16	66	92	69	29	5	1	0
	CDD10.0	3509	377	339	368	321	268	186	167	191	239	321	349	383
	CDD18.3	747	119	106	109	72	26	2	1	2	18	67	100	125
	CDH23.3	3975	512	447	464	319	119	17	8	39	196	541	643	671
	CDH26.7	690	79	66	61	29	5	0	0	2	26	132	156	134
Wind	WSAvg	2.6	2.6	2.6	2.6	2.4	2.2	2.4	2.7	2.9	3.2	3.1	2.7	2.4
Precipitation	PrecAvg	1414	301	270	195	52	19	5	8	7	8	61	159	305
	PrecMax	1840	796	632	508	157	156	34	38	31	61	311	362	720
	PrecMin	840	78	31	30	0	0	0	0	0	0	1	1	54
	PrecStd	268	165	138	102	44	28	7	10	7	11	57	80	143
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.9	29.8	29.2	28.9	27.2	25.1	24.2	26.2	29.0	31.2	31.2	30.8
		MCWB	20.9	21.5	21.0	19.6	18.6	16.7	15.8	15.3	16.8	18.5	19.2	20.5
	2%	DB	28.4	28.2	28.1	27.5	26.0	23.5	22.8	24.2	27.1	29.7	29.9	29.2
		MCWB	20.8	21.0	21.0	19.4	18.1	16.1	14.8	15.3	16.2	18.2	19.0	20.2
	5%	DB	27.3	27.2	27.1	26.2	24.8	22.2	21.3	22.9	25.8	28.1	28.6	28.1
		MCWB	20.7	20.7	20.5	19.2	17.7	15.9	14.7	14.8	16.1	17.7	19.0	20.3
	10%	DB	26.2	26.1	26.1	25.2	23.2	20.9	20.1	21.2	24.0	26.4	27.1	27.0
		MCWB	20.5	20.4	20.2	19.0	17.4	15.5	14.4	14.6	15.8	17.3	18.8	20.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.9	23.8	23.6	22.4	21.0	19.4	18.2	18.2	19.3	21.6	22.9	23.9
		MCDB	26.7	26.7	27.0	25.8	24.7	22.4	21.0	22.1	25.0	27.5	27.1	27.5
	2%	WB	23.0	22.7	22.6	21.2	19.7	17.9	16.7	16.7	18.1	20.2	21.6	22.7
		MCDB	26.0	26.2	26.1	24.8	23.2	21.1	20.2	21.4	23.6	25.4	26.4	26.5
	5%	WB	22.0	21.8	21.7	20.4	18.9	17.0	15.7	16.0	17.3	19.3	20.7	21.7
		MCDB	25.3	25.5	25.4	24.2	22.5	20.1	19.6	20.7	22.8	24.7	25.4	25.7
	10%	WB	21.2	21.0	20.9	19.6	17.9	16.1	14.9	15.2	16.5	18.3	19.8	20.8
		MCDB	24.6	24.7	24.3	23.5	21.7	19.5	18.7	19.6	21.8	23.4	24.3	24.8
Mean Daily Temperature Range	5% DB	MDBR	9.5	9.4	9.5	10.2	10.3	10.4	10.5	11.3	12.6	13.1	12.2	10.9
		MCDBR	11.0	10.8	10.8	11.6	12.2	12.2	12.3	13.6	15.3	15.3	14.0	12.1
		MCWBR	5.2	5.2	5.2	5.3	5.7	6.1	6.1	6.0	6.1	5.7	5.4	5.3
	5% WB	MCDBR	9.6	9.8	10.1	10.6	10.7	10.9	11.0	11.7	13.1	13.3	12.1	11.0
		MCWBR	5.6	5.6	5.7	5.8	6.0	6.5	6.3	6.4	6.3	6.1	5.9	5.9
Clear-Sky Solar Irradiance	taub	0.333	0.327	0.318	0.299	0.284	0.268	0.272	0.285	0.316	0.363	0.369	0.345	
	taud	2.668	2.702	2.730	2.780	2.794	2.803	2.736	2.630	2.500	2.376	2.416	2.562	
	Ebn at Noon	1011	1006	993	977	959	959	962	976	978	958	968	998	
	Edh at Noon	98	93	87	78	71	68	74	88	108	128	125	109	
All-Sky Solar Radiation	RadAvg	5.83	5.79	5.58	5.23	4.43	4.04	4.12	4.94	6.03	6.54	6.47	6.39	
	RadStd	0.56	0.49	0.26	0.27	0.21	0.18	0.22	0.29	0.33	0.34	0.38	0.43	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+0.29	N/A	N/A	+0.48	+1.40	+1.66	N/A	-62	+164	+82
Regional (0 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page